MATH 152
Mrs. Bonny Tighe

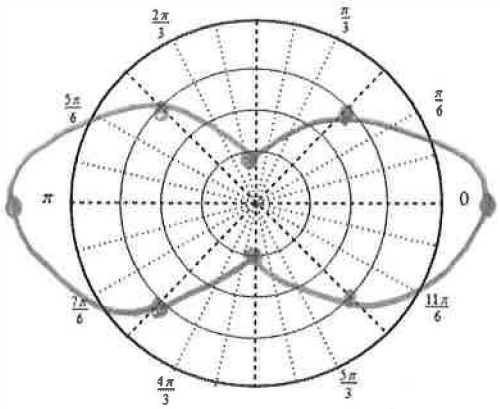
QUIZ 11A

25 points
10.2, 10.3

NAME Solutions

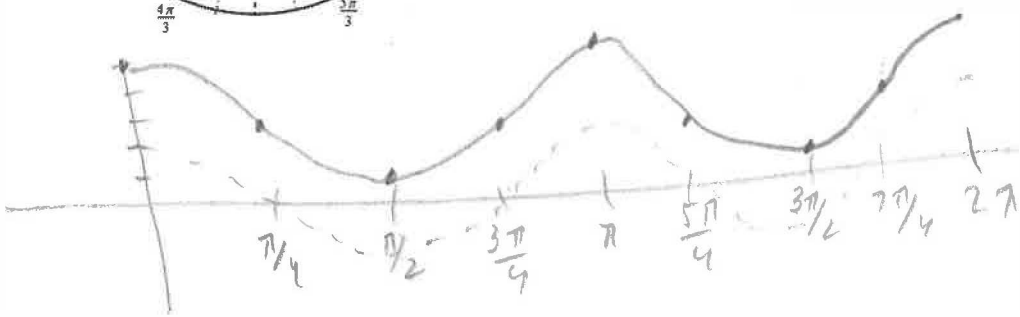
TA_____ Wed 12/07/2022

1. Sketch the graph. $r = 3 + 2\cos 2\theta$.



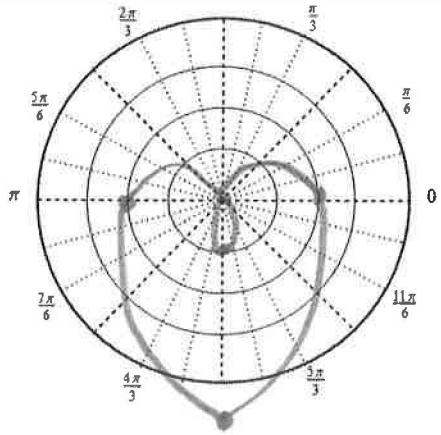
"peanut"

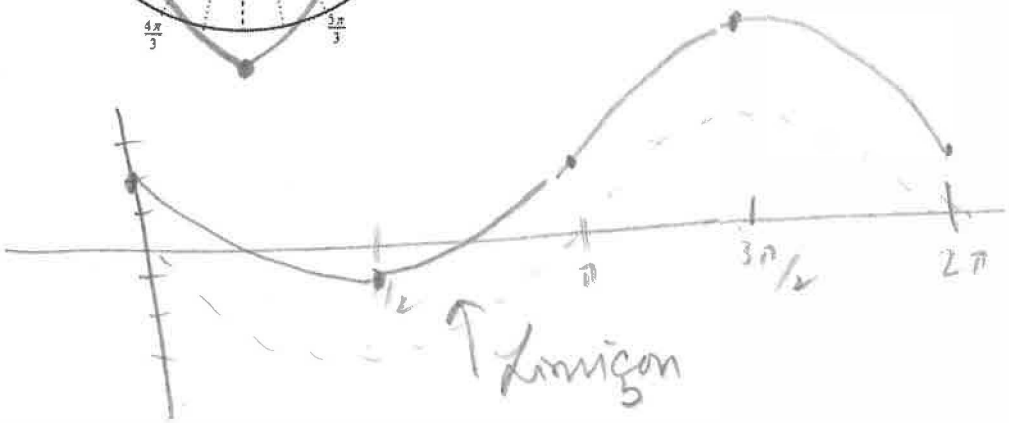
5pts



2. Sketch the graph $r = 2 - 3\sin\theta$

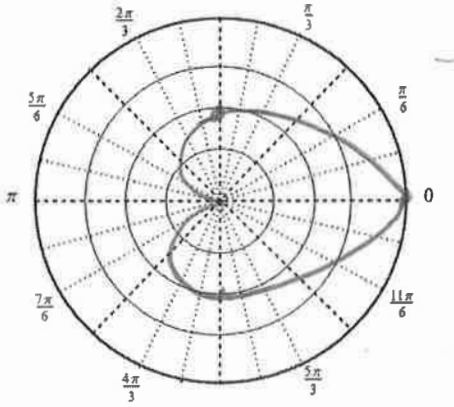
5pts





3. Sketch the graph $r = 2 + 2\cos\theta$

5pts



4. a) Find the Cartesian coordinates for the polar coordinate $(-2, -5\pi/6)$

$(1, \sqrt{3})$

2pts

$x = r\cos\theta = -2\cos(-5\pi/6) = -2(-\sqrt{3}/2) = \sqrt{3}$

$y = r\sin\theta = -2\sin(-5\pi/6) = -2(-1/2) = 1$

b) Find polar coordinates for the Cartesian coordinate $(3, -3\sqrt{3})$ where $r < 0$, *and* $\theta > 0$

2pts

$r^2 = x^2 + y^2 = r^2 = 9 + 9(3) = 36$ $\quad r = 6$

$\tan\theta = -3\sqrt{3}/3 = -\sqrt{3}$ $\quad \theta = -\pi/3$

$\cancel{(6, 5\pi/3)}$

$(-6, 2\pi/3)$

c) Give three alternate versions for the polar point $(4, 5\pi/4)$

3pts

$r > 0, \theta < 0$ $(4, -3\pi/4)$

$r < 0, \theta < 0$ $(-4, -7\pi/4)$

$r < 0, \theta > 0$ $(-4, \pi/4)$

5. How may petals would the graph of each of the following have?

3pts

a) $r = 3\sin 4\theta$ 8

b) $r = -4\cos 7\theta$ 7

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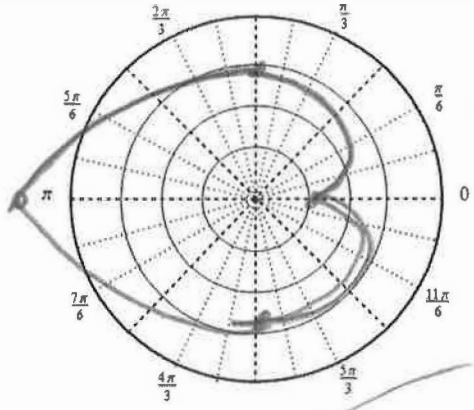
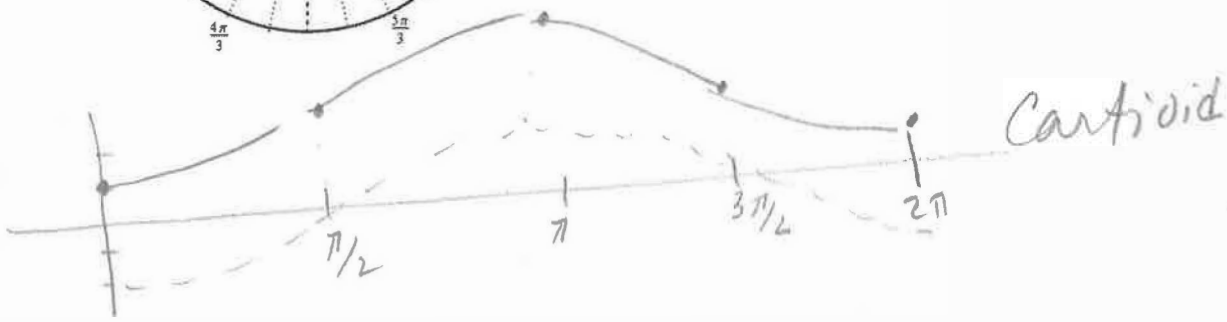
1. Sketch the graph. $r = 3 - 2\cos\theta$.

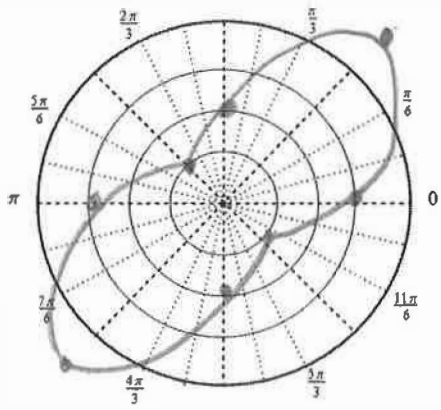
5pts



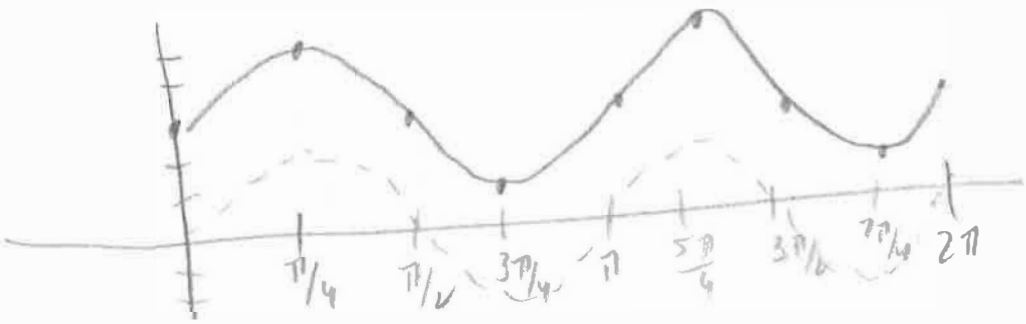


5pts

2. Sketch the graph $r = 3 + 2\sin 2\theta$



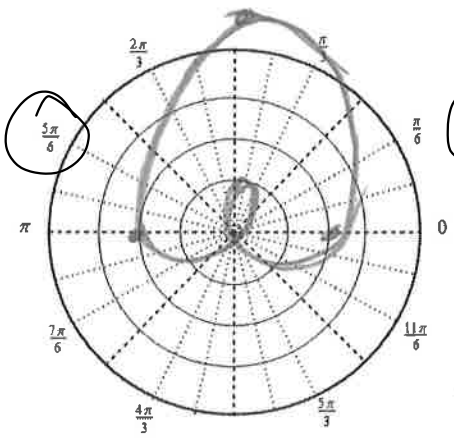
"peanut"

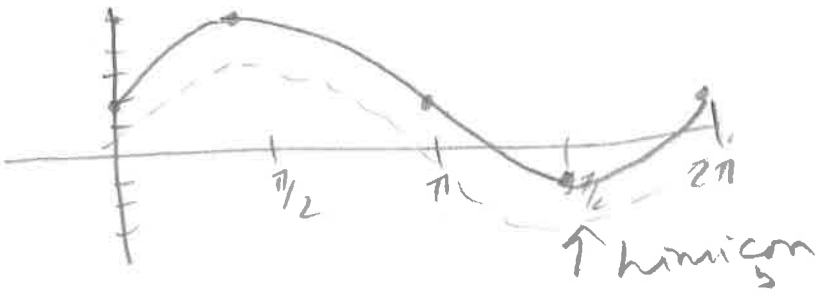


5pts

3. Sketch the graph $r = 2+3\sin\theta$

$\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$





2pts

4. a) Find the Cartesian coordinates for the polar coordinate $(3, -7\pi/6)$

$x = r\cos\theta = 3(\cos -7\pi/6) = 3(-\sqrt{3}/2) = -3\sqrt{3}/2$

$y = r\sin\theta = 3\sin(-7\pi/6) = 3(1/2) = 3/2$

$\left(-\frac{3\sqrt{3}}{2}, 3/2\right)$

2pts

b) Find polar coordinates for the Cartesian coordinate $(-\sqrt{3}, -1)$ where $r > 0$, and $\theta > 0$

$r^2 = x^2 + y^2$

$r^2 = 3+1 = 4 \quad r = 2$

$\tan\theta = -1/-\sqrt{3} = 1/\sqrt{3}$ $\quad \theta = \pi/6$ or $7\pi/6$

$(2, 7\pi/6)$

3pts

c) Give three alternate versions for the polar point $(2, 5\pi/3)$

$r > 0, \theta < 0$ $(2, -\pi/3)$

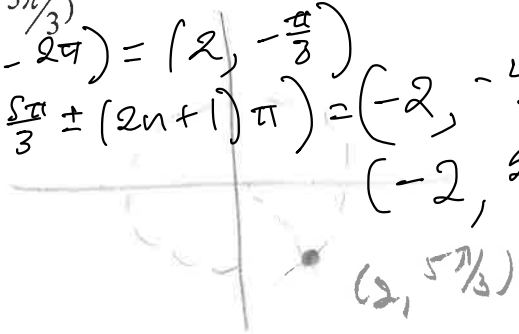
$r < 0, \theta < 0$ $(-2, -4\pi/3)$

$r < 0, \theta > 0$ $(-2, 2\pi/3)$

$\left(2, \frac{5\pi}{3} - 2\pi\right) = \left(2, -\frac{\pi}{3}\right)$

$\left(-2, \frac{5\pi}{3} \pm (2n+1)\pi\right) = \left(-2, -\frac{4\pi}{3}\right)$

$\left(-2, \frac{2\pi}{3}\right)$



3pts

5. How may petals would the graph of each of the following have?

a) $r = 3\sin 5\theta$ 5

b) $r = -4\cos 6\theta$ 12